
Project Report

Precision Closed-Loop AC Power Controller using PID

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Course Code: PH3144

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1 Introduction

In any modern electric system, especially in labs, precise control over the power delivered to an electrical appliance/device is important for efficiency, stability and protection of devices. What we've tried to do is to precisely control the electrical power of any device. We are using a simple light bulb for the ease of demonstration. But in theory, we can implement this on any device connected to AC Mains.

Our device measures the current and voltage input for an electric load and allows us to fix a power in the working range of the appliance. We control this with a technique called *phase-angle fired control*, where the firing angle of a thyristor (Triac) is adjusted to control the portion of the AC Waveform delivered to the load.

For an electric bulb, this works as a dimmer for the system, allowing us to observe the difference. But a simple dimmer is an open-loop control, whilst we have implemented a closed-loop approach with help of *PID*. You cannot ask the dimmer to set the bulb to run at 50 W. You can only set it to 50% effort, but you can't be sure. It is also susceptible to disturbances. If there is an increase in the voltage in the circuit, the dimmer doesn't know that, and the bulb will light brighter.

A closed-loop control is better because it uses the feedback to compare the result. We are constantly measuring the current and voltage to find the actual Power and compare it to the set-point to dynamically adjust the firing angle based on the feedback.

2 Design

2.1 Block Diagram

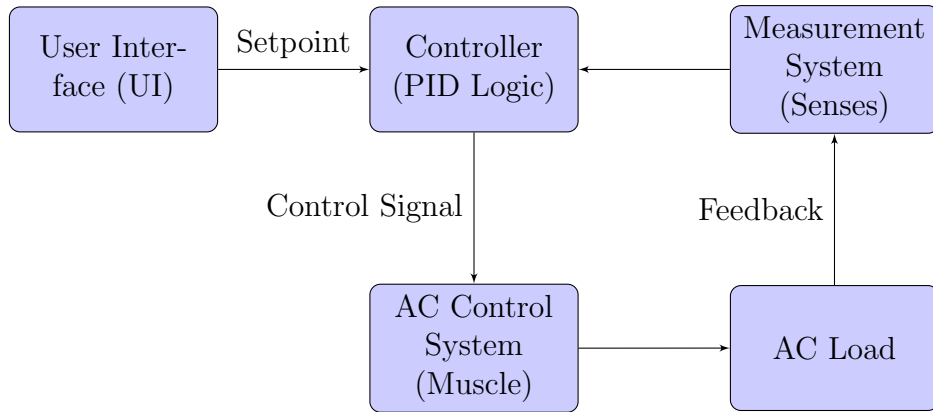


Figure 1: Block Overview of the Closed-Loop Power Controller

We divide our system into smaller task-specific parts to better understand our system. First is the User Interface (UI), it allows us to display the results and also fix a set-point for the system. The Controller (here Arduino) is the brain of the whole system, running the PID logic, and controlling all the other components of our project. The Measurement System consists of some simple sensors that allow us to take current and voltage measurements and send these to the Controller. The AC Control is the muscle of the system, it controls the parameters to maintain the set-point as dictated by the Controller.

2.2 Hardware Design

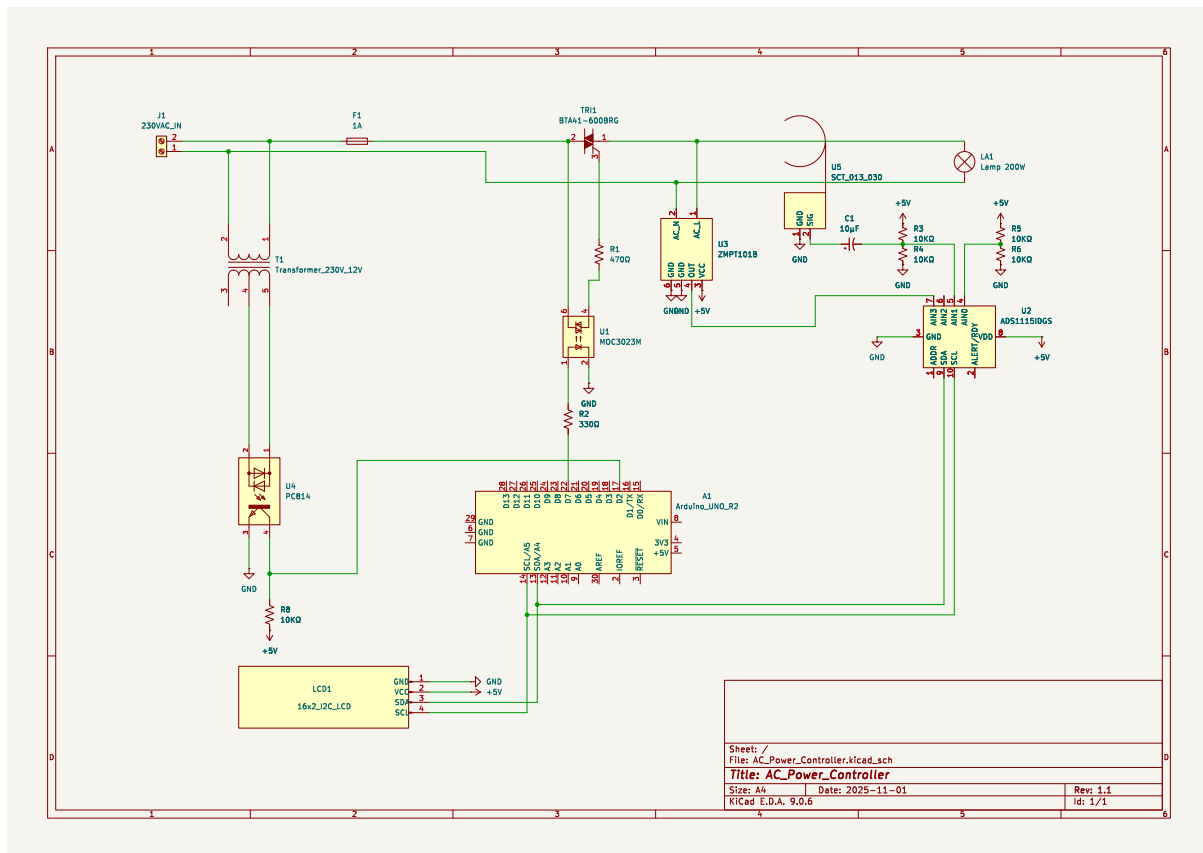


Figure 2: KiCad Schematics for the whole working the circuit

2.2.1 Measurement Module

This module is responsible for providing the critical feedback to the controller. The major components of the measurement module are:

1. ZMPT101B Voltage Sensor Module

This active module safely steps down the 230V AC mains voltage to a low-voltage, isolated AC sine wave. Its onboard op-amp circuit biases this signal around a 2.5V DC offset, making it safe for the low-voltage components to read.

2. SCT-013-030 Current Sensor

A non-invasive current transformer that clamps around the live wire. It outputs a small AC voltage (0-1V) that is directly proportional to the AC current flowing to the load. This signal is also biased at 2.5V using an external voltage divider.

3. ADS1115 16-bit ADC Module

This is a high-precision, 16-bit Analog-to-Digital Converter. It is connected to the Arduino via the I²C bus (A4/A5). The Arduino's built-in 10-bit ADC is too noisy and has insufficient resolution for a "precision" controller. The 16-bit ADS1115 provides high-accuracy, low-noise measurements.

2.2.2 AC Control Module (Muscle)

This component is the high-voltage part of the circuit that controls the power flow. The major components that make up the AC Control Module are:

1. MOC3023M (Triac Driver Optocoupler)

This chip is the safe bridge between the Arduino and the Triac. It receives a low-voltage 5V "fire" command from the Arduino (on pin D7) and uses an internal light-activated switch to trigger the high-voltage Triac.

This provides galvanic isolation. It is the most important safety component, creating a physical air gap between the 230V AC circuit and the 5V logic circuit, protecting the Arduino and the user from lethal voltage.

2. BTA41600B (Triac)

This is the heavy-duty, high-power AC switch. When it receives the trigger signal from the MOC3023M, it "latches" ON and conducts current to the light bulb. It automatically turns itself OFF (resets) at the end of every AC half-cycle when the current drops to zero.

This is the component that actually controls the power. By precisely timing when this switch is turned on within each 10ms half-cycle, we can control the total energy delivered to the load.

3. PC814 (AC-Input Optocoupler)

This chip functions as our Zero-Crossing Detector (ZCD). Its AC input side watches the sine wave from the step-down transformer. At the precise moment the wave crosses its center point, the optocoupler's internal LED turns off, and triggers the phototransistor on the output side. This interrupt signal pulls down the output PIN_4 that is connected to +5V of the Arduino.

This signal is connected to the Arduino's hardware interrupt pin (D2). It provides a perfect timing pulse 100 times per second, synchronizing the Arduino's code with the AC line.

3 Methodology

4 Implementation

5 Software

6 Debugging

7 Results & Discussion